



الرسم بالحاسب

12:00-2:00

18/6/2023

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Faculty of Computers & Information, Assiut University

Course code: CS351

3rd Level

Final Exam

Duration: 2 hours

Total Marks: 50

Reviewer: (Add the name)

This exam for the following program(s) : (Add the program's name(s))

1

الإسم الرباعي (بالعربي فقط) *

علي هاشم محروس عبدالحافظ

2

رقم الجلوس *

1620195167

3

المستوي *

☐ الاول

☐ الثاني

☐ الثالث

☐ الرابع 2013

☐ الرابع 2014

☐ الرابع 2015

☐ الرابع 2016

☐ الرابع 2017

☐ الرابع 2018

☒ الرابع 2019

4

» * البرنامج

☐ عام

☒ بايو

☐ هندسة

☐ ذكاء صناعي

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رقم المعمل *



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6

رقم الكمبيوتر * 

8

7

الكود * (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

NRCI

 **Will be reviewed**

8

What is the difference between glColor3d and glColor3f? 

- ☐ glColor3d only sets RGB, while glColor3f sets R, G, B and A
- ☐ glColor3d allows 3d color operations, while glColor3f only allows 8-bit
- ☒ glColor3d takes double arguments, while glColor3f takes float arguments
- ☐ glColor3d is in real space, glColor3f is in integer space

 **Will be reviewed**

9

This 3D homogeneous point: (4,3,2,1) corresponds to what Cartesian point: 

- ☐ (4,3,2,1)
- ☐ (3,2,1)
- ☐ (3/4, 1/2, 1/4)
- ☒ (4,3,2)

 **Will be reviewed**

10

The function `glPolygonMode` allows polygons to be rendered as filled solids, as outlines, or as points. 

- ☒ True
- ☐ False

 **Will be reviewed**

11

Double buffering solves which problem: 

- ☐ producing stereo images
- ☐ remote debugging PCs
- ☒ displaying a frame buffer while it is being updated.
- ☐ Enhancing performance

 **Will be reviewed**

12

To map the output of the OpenGL code to area A (as shown in the figure), we use the 

- ☒ `glViewport(0, 0, 125, 125);`
- ☐ `glViewport(125, 125, 125, 125);`
- ☐ `glViewport(125, 125, 0, 0);`
- ☐ `glViewport(0, 125, 0, 125);`

📝 **Will be reviewed**

13

..... light component comes from one direction and once it hits a surface, it is scattered. 📎

- ☐ Diffuse
- ☒ Specular
- ☐ Ambient
- ☐ Spot

📝 **Will be reviewed**

14

Does OpenGL provide modeling primitives such as a cube, cone, pyramid? 📎

- ☐ Not in the official specification, but such functionality is provided by the GLUT library.
- ☒ Yes, but only with `glEnable(GL_PRIMITIVES)` called before the `glBegin` statement.
- ☐ Yes.

☐ In the official specification, but also enhanced versions provided by the GLUT library.

 **Will be reviewed**

15

Whenever the window size changes, the viewport and clipping volume must be redefined. 

☒ True

☐ False

 **Will be reviewed**

16

The process of removal of hidden surfaces is termed as 

☐ clipping

☐ copying

☒ culling

☐ shorting

📝 **Will be reviewed**

17

What is the coordinates of point $p1=(2,2)$ after executing the following OpenGL code segment:

```
glTranslate(2, 0, 0);  
glScalef(1.0, -1.0, 1.0);  
glRotate(180, 0, 0, 1);  
drawPoint_p1();
```

☐ (0,-2)

☐ (2,2)

☐ (2,0)

☒ (0,2)

📝 **Will be reviewed**

18

One important rule when you use quads is that all four corners of the quadrilateral must lie in

a plane. 

☒ True

☐ False

 **Will be reviewed**

19

The dot product of any two perpendicular vectors equals one 

☐ True

☒ False

 **Will be reviewed**

20

glOrtho is used to define a _____ matrix. 

☐ model view

☐ projection

☒ view

☐ model

 **Will be reviewed**

21

OpenGL can draw convex polygons. 

☒ True

☐ False

 **Will be reviewed**

22

The following two code segments are equivalent:

Code#1

```
glScalef(1.0, 1.0, 2.0);  
glScalef(1.0, 1.0, 4.0);  
drawShape_3();
```

Code#2

```
glScalef(1.0, 1.0, 4.0);  
glScalef(1.0, 1.0, 2.0);  
dawShape_3();
```

☐ True

☒ False

 **Will be reviewed**

23

Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glRotatef(90.0, 0, 0, 1.0);  
glPushMatrix();  
glTranslatef(-1.0f, -2.0f, -3.0f);  
glRotatef(-90.0, 0, 0, 1.0);  
glScalef (-1.0,0.5, -1.0);  
glPopMatrix();  
glRotatef(90.0, 0, 0, 1.0);
```

☐ A

☐ B

☒ C

☐ D

📝 **Will be reviewed**

24

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINES);  
    glVertex2fv(A);  
    glVertex2fv(B);  
    glVertex2fv(C);  
    glVertex2fv(D);  
    glVertex2fv(E);  
    glVertex2fv(F);  
glEnd();
```

☒ A

☐ B

☐ C

☐ D

 **Will be reviewed**

25

Once a 2D or 3D rotation is performed, is it possible to reverse the rotation? 

☒ Yes

☐ No

 **Will be reviewed**

26

Two translation matrices, T_1 and T_2 can be combined into another translation matrix T by multiplying them, i.e. $T = T_1 T_2$ 

☐ True

☒ False

 **Will be reviewed**

27

Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glRotatef(90.0, 0, 0, 1.0);  
glScalef(-1.0, 2.0, -1.0);
```

☐ A

☐ B

☒ C

☐ D

 **Will be reviewed**

28

Does OpenGL use column-major order or row-major order when representing a transformation matrix? 

☐ row-major

☒ column-major

📝 **Will be reviewed**

29

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_QUAD_STRIP);  
    glVertex2fv(v1); glVertex2fv(v2);  
    glVertex2fv(v3); glVertex2fv(v4);  
    glVertex2fv(v5); glVertex2fv(v6);  
    glVertex2fv(v7); glVertex2fv(v8);  
glEnd();
```

☐ A

☐ B

☒ C

☐ D

📝 **Will be reviewed**

30

What is the significance of i,f,v,d... in function naming conventions found in OpenGL? 📝

- ☐ OpenGL's naming conventions vary and must be examined in the specification for the version you are using
- ☒ Specify the expected parameter types (int, float, pointer, double, etc.)
- ☐ Specify the target type for variables written by OpenGL on the CPU-side only
- ☐ Specify the output target (immediate mode, frame buffer, virtual mode, double-buffer, etc.)

 **Will be reviewed**

31

For a rotation matrix R and a translation matrix T, $RT = TR$, even though matrix multiplication is not generally commutative 

- ☐ True
- ☒ False

 **Will be reviewed**

32

$\text{Rotate}(z, 90) * \text{Rotate}(z, 45) = \text{Rotate}(z, 45) * \text{Rotate}(z, 90)$ 

- ☒ True

☐ False

 **Will be reviewed**

33

OpenGL have functions and commands relating to window or screen management 

☒ True

☐ False

 **Will be reviewed**

34

What is the correct order of operations of the OpenGL graphics pipeline? 

☐ projection transformation, modelview transformation, divide by w, viewport transform

☒ modelview transformation, divide by w, projection transformation, viewport transform

☐ modelview transformation, viewport transform, divide by w, projection transformation

☐ modelview transformation, projection transformation, divide by w, viewport transform

 **Will be reviewed**

35

The OpenGL command `glClearColor(0.0f, 1.0f, 0.0f, 1.0f)` clear a window with blue color 

☐ True

☒ False

 **Will be reviewed**

36

Generally, what primitive polygon is used for creating a mesh to represent a complex object?



☒ Triangle

☐ Cube

☐ Sphere

☐ Circle

 **Will be reviewed**

37

By default, in OpenGL, polygons that have counterclockwise winding are considered back facing. 

☐ True

☒ False

 **Will be reviewed**

38

Can glNormal be called between glBegin and glEnd ? 

☒ True

☐ False

 **Will be reviewed**

39

An example of a transformation matrix for nonuniform scale is 

☐ A

☐ B

☐ C

☒ D

 **Will be reviewed**

40

OpenGL runs only on Microsoft operating systems. 

☐ True

☒ False

 **Will be reviewed**

41

Rotating the 3D point $(0, -1, 0)$ by -90 around z -axis gives what: 

- ☒ $(1, 0, 0)$
- ☐ $(-1, 0, 0)$
- ☐ $(0, 1, 0)$
- ☐ $(0, -1, 0)$
- ☐ $(-1, -1, 0)$

 **Will be reviewed**

42

The product of several rotation matrices is itself a rotation. 

- ☒ True
- ☐ False

 **Will be reviewed**

43

Is it possible to perform a reflection of an object across a specified axis? 

- ☐ No, this functionality is not possible with the current version of OpenGL
- ☐ Yes, with `glReflect()`
- ☐ No, you must redraw your object reflected across the desired axis
- ☒ Yes with simple matrix multiplication

 **Will be reviewed**

44

What does `glClear` do? 

- ☐ It clears all OpenGL buffers
- ☒ It clears the OpenGL buffers specified in an argument
- ☐ It sets the window color to black

☐ None of the above

 **Will be reviewed**

45

..... * Rotate(90.0, 0, 0, 1.0) = Rotate(90.0, 0, 0, 1.0)* Translate(2,-1,3) 

☐ Translate(1,2,3)

☐ Translate(1,3,-2)

☐ Translate(-2,1,3)

☒ Translate(2,-1,3)

 **Will be reviewed**

46

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_TRIANGLE_STRIP);  
    glVertex2fv(v1); glVertex2fv(v2);  
    glVertex2fv(v3); glVertex2fv(v4);  
    glVertex2fv(v5); glVertex2fv(v6);
```

```
glVertex2fv(v7); glVertex2fv(v8);  
glEnd();
```

☐ A

☒ B

☐ C

☐ D

 **Will be reviewed**

47

The RGB of A cyan color is

☐ (1.0,1.0,0.0)

☐ (1.0,0.0,1.0)

☒ (0.0,1.0,1.0)

☐ (1.0,1.0,1.0)

 **Will be reviewed**

48

For nonzero vectors u and v , $u \times v = \dots\dots\dots$, if and only if u and v are parallel 

- ☒ (0 , 0 , 0)
- ☐ (1 , 0 , 0)
- ☐ (0 , 1 , 0)
- ☐ (0 , 0 , 1)
- ☐ (1 , 1 , 1)

 **Will be reviewed**

49

Positive value for the rotation angle Θ defines. 

- ☒ Clockwise rotation about the pivot point
- ☐ Counterclockwise rotation about the pivot point

📝 **Will be reviewed**

50

Which of the following is not a valid argument to glBegin ()? 📖

- ☒ GL_POINTS
- ☐ GL_LINES
- ☐ GL_LINE_STRIP
- ☐ GL_TRIANGLES
- ☐ GL_QUADS_FAN

📝 **Will be reviewed**

51

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_LOOP);  
    glVertex2fv(A);  
    glVertex2fv(B);  
    glVertex2fv(C);  
    glVertex2fv(D);  
    glVertex2fv(E);
```

```
glVertex2fv(F);  
glEnd();
```

☐ A

☐ B

☒ C

☐ D

 **Will be reviewed**

52

If the scaling factor value is less than one, then

☒ It reduces the size of object

☐ It increases the size of object

☐ It does not change the shape of an object

☐ None

📝 **Will be reviewed**

53

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glRotatef(45.0, 0, 0, 1.0);  
glRotatef(45.0, 0, 0, 1.0);
```

☐ A

☐ B

☒ C

☐ D

📝 **Will be reviewed**

54

Which of the following is not a valid argument to glMatrixMode? 

☐ GL_PROJECTION

☒ GL_DEPTH

☐ GL_MODELVIEW

📝 **Will be reviewed**

55

The OpenGL command `glClearColor(0.0f, 1.0f, 0.0f, 1.0f)` clear a window with blue color 📋

☐ True

☒ False

📝 **Will be reviewed**

56

What is the normalized surface normal vector to the plane $x+2y+2z+4=0$? 📋

☐ (1,2,2)

☐ $(1/3, 2/3, 2/3, 4/3)$

☐ $(1/3, 2/3, 2/3, -4/3)$

 **Will be reviewed**

57

Is the following code correct?

```
glBegin (GL_QUADS);  
glVertex2i (1, 0);  
glVertex2i (1, 1);  
glVertex2i (-1, 1);  
glVertex2i (-1, 0);  
glEnd (GL_QUADS);
```



☐ Yes

☒ No

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